

Tools For Troubleshooting Theory

Task 1: Three Real-Life Multimeter Troubleshooting Scenarios

1. Testing a PC Power Supply Unit (PSU):

- *How it helps:* It measures whether the 12V, 5V, and 3.3V output pins on the 24-pin ATX connector deliver correct, stable voltages to rule out a dead or failing power supply.

2. Checking Motherboard / Smartphone Charging Board Continuity:

- *How it helps:* Using continuity/diode mode, it beeps to confirm if a copper trace or charging port connection is broken or shorted to ground.

3. Testing a Motherboard CMOS Coin-Cell Battery:

- *How it helps:* It reads the exact DC voltage across the CR2032 battery terminals to verify if it has dropped below its required 3V threshold, which causes BIOS time/setting resets.

Task 2: POST Card YouTube Demo Summary (3–4 Lines)

- In the demonstration, a diagnostic POST card plugged into the motherboard's PCIe/PCI slot displays cycling two-digit hexadecimal codes representing Power-On Self-Test initialization stages.
- As the motherboard boots, the digital display quickly changes through system checks until it halts and freezes on code **55** (or **C1** depending on the BIOS vendor).
- According to the motherboard's debug code table, this specific halted code signifies that memory initialization failed.
- This allows the technician to immediately identify an unseated or faulty RAM stick as the root cause of the blank screen without guessing.

Task 3: Free Software Diagnostic Tool Guide

- **Recommended Tool:** CrystalDiskInfo (or HWMonitor / Speccy).
- **Steps to Complete:**
 1. Download and run the portable version of **CrystalDiskInfo** or **HWMonitor**.
 2. Open the main window showing overall health status (e.g., storage drive health percentage, temperature, or CPU thermal readings).
 3. Press **Win + Shift + S** (Snipping Tool) to take a screenshot.
 4. **Important Constraint:** Crop out or blur any sensitive identifiers, such as serial numbers, drive letters containing personal folder names, or PC usernames, before attaching it to your submission.

Task 4: Comparison Table (Multimeter vs. POST Card vs. Software Diagnostics)

Troubleshooting Tools: Comparison & Applications		
Day 1 Task 4: Multimeter vs. POST Diagnostic Card vs. Software Diagnostics		
Tool	What It Checks	Example Problem Detected
Digital Multimeter Hardware Electrical Testing	Direct physical electrical properties across components: - DC / AC Voltage levels - Circuit Continuity (detecting breaks / shorts) - Resistance (Ohms) across coils and traces - Diode and power rail integrity	Power Supply / Motherboard Power Rail Failure - A PC powers off unexpectedly because the PSU +12V rail sags to 9.8V under load. - A motherboard fails to boot due to a shorted ceramic capacitor on the 19V DC input line. - A dead CMOS CR2032 battery measuring below 2.5V (nominal 3.0V), resetting BIOS settings.
POST Diagnostic Card Motherboard Startup & Bus Signals	BIOS initialization stages and bus communications during the Power-On Self-Test (POST) - Tests CPU, RAM, and GPU sequencing - Clock signals (CLK) and Reset line status 2-digit / 4-digit hexadecimal POST debug codes	POST Error Codes / No Display (Beep Codes) - System powers on with spinning fans but no video signal, halting on hex code '55' or 'C1' to indicate unseated or defective RAM modules. - Reset LED stays permanently lit, pointing to a dead CPU or corrupt BIOS chip.
Software Diagnostics (e.g., CrystalDiskInfo, HWMonitor) Live System Health & Component Sensors	Operating metrics and telemetry while the OS or live boot environment is running - Storage S.M.A.R.T. health and bad sectors - Real-time CPU / GPU thermal sensors - Fan RPMs and internal VRM rail voltages - RAM error checking (MemTest86)	System Configuration & Thermal Throttling - CrystalDiskInfo flags a 'Caution' alert with reallocated and pending bad sectors on an NVMe SSD, preventing data loss before drive failure. - HWMonitor shows the CPU running at 98°C with thermal throttling caused by dried thermal paste or a failing AIO pump.